Applied Statistics, an introduction

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The scientific question

A running example

We want to know the **mean diameter of the trees in a forest**.

The population

- all trees in the forest of interest;
- each tree has a diameter;
- the population mean is

$$\mu = \text{mean diameter of all trees.}$$

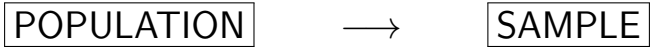
The practical problem

- the forest may contain thousands or millions of trees;
- measuring every tree may be impossible or too expensive;
- we therefore observe a sample.

The statistical problem

How can information from a sample tell us something about the population mean μ?

Population and sample: keep them separate



Population

The complete set of units about which we want to make a statement.

$$\mu = \text{population mean}$$

μ is a **fixed but usually unknown quantity**.

Sample

The units actually selected and measured.

$$\bar{x} = \text{observed sample mean}$$

$\bar{x}$ is a **number calculated from the data**.

A rule worth remembering

Population $\neq$ sample. Population parameter $\neq$ sample statistic.

A population can be fixed while the sample changes

Imagine that the forest is fixed.

$$\text{same population} \implies \begin{matrix}\text{sample 1}\\ \text{sample 2}\\ \text{sample 3}\end{matrix}$$

$$\bar{x}_1 = 28.4 \qquad \bar{x}_2 = 29.7 \qquad \bar{x}_3 = 27.9$$

The population has not changed.
What changed is **which trees were selected**.

This is the source of sampling uncertainty

Different samples from the same population generally produce different sample statistics.

Before sampling and after sampling

This distinction is essential.

Before the sample is selected

We do not know which trees will be selected. Therefore

$$\bar{X}$$

is a **random variable**.

After the sample is observed

The selected trees are known.
We have one numerical result:

$$\bar{x}$$

It is no longer random in this experiment.

Notation matters

$\bar{X}$ = random sample mean

$\bar{x}$ = observed sample mean

Why is $\bar{X}$ random?

Suppose we repeatedly draw samples of $n = 20$ trees using the same sampling procedure.

sample	sample mean
1	28.4
2	29.7
3	27.9
4	30.1
⋮	⋮
1000	28.8

The value of $\bar{X}$ depends on the random sample selected.

$$\boxed{\text{random sample} \longrightarrow \text{random value of } \bar{X}}$$

The fact that we eventually observe one value does **not** mean that the variable was not random before observation.

Probability: a language for sampling variability

Probability becomes useful because we can ask:

- How often is $\bar{X}$ close to μ?
- How variable is $\bar{X}$ from one sample to another?
- What values of $\bar{X}$ are plausible?

This requires describing the probability distribution of $\bar{X}$.

The bridge

$$\boxed{\text{sampling procedure}} \longrightarrow \boxed{\text{random variable } \bar{X}} \longrightarrow \boxed{\text{distribution of } \bar{X}}$$

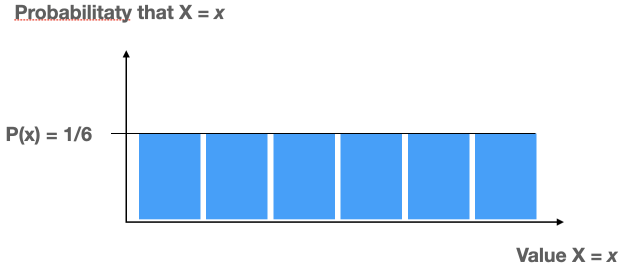
That distribution is called the **sampling distribution** of the mean.

A small probability example

Before considering tree diameters, consider a simple experiment.
Throw a fair die:

$$X \in \{1, 2, 3, 4, 5, 6\}, \quad \text{where } P(X = x) = \frac{1}{6}.$$

The outcome is random, but the probability distribution is known.



The important analogy

For a die, we know the distribution of X.

Random variables

A random variable is a numerical quantity whose value is determined by a random experiment. More formally:

$$X : \Omega \to \mathbb{R}.$$

For our forest example, the individual tree diameter can be represented by a random variable X when we select a tree according to the sampling mechanism.
For a sample of n trees:

$$X_1, X_2, \ldots, X_n$$

represent the observed values associated with the selected units.

Keep the levels distinct

X_i = one observation.
$\bar{X}$ = a statistic calculated from the whole sample.
μ = a population parameter.

Discrete and continuous random variables

Discrete

Possible values are finite or countably infinite.
Examples:

- number of trees;
- number of diseased plants;
- number of seedlings.

Continuous

Possible values form a continuum.
Examples:

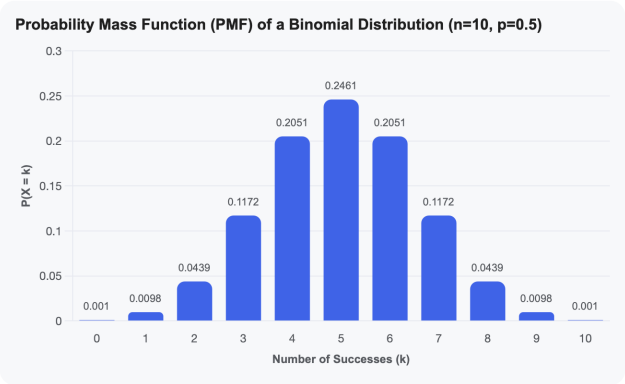
- diameter;
- height;
- temperature;
- biomass.

The distinction is mathematical: it concerns the set of possible values and how probabilities are represented.

Distributions: what values are plausible?

A probability distribution describes how probability is allocated over possible values.
For a discrete variable:

$$p(x) = P(X = x), \qquad \sum_x p(x) = 1.$$



Continuous variables: density is not probability

For a continuous random variable, the probability of hitting an individual point is zero:

$$P(X = x) = 0.$$

Instead, we look at intervals:

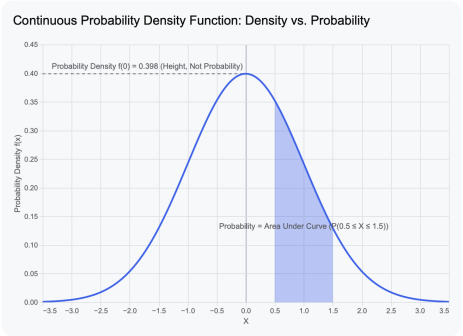
$$P(a < X < b) = \int_a^b f(x)\,dx.$$

The probability density function (PDF) satisfies:

$$f(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{+\infty} f(x)\,dx = 1.$$

Continuous variables: density is not probability

$$P(a < X < b) = \int_a^b f(x)\,dx.$$



Think "area", not "height"

$f(x)$ is a density. Probability is obtained by integrating it.

Expected value: the centre of a distribution

For a discrete random variable:

$$E(X) = \sum_x xP(X = x).$$

For a continuous random variable:

$$E(X) = \int_{-\infty}^{+\infty} xf(x)\,dx.$$

We often write

$$\mu = E(X).$$

Interpretation

The expected value is the long-run average value produced by the probability model.

Important: μ is a population/distribution characteristic; $\bar{x}$ is a sample result.

Variance and standard deviation

The variance describes the spread of a random variable:

$$\mathrm{Var}(X) = E[(X - \mu)^2].$$

The standard deviation is

$$\sigma = \sqrt{\mathrm{Var}(X)}.$$

$\boxed{\mu = \text{centre}}$ $\boxed{\sigma = \text{scale of variation}}$

For tree diameters, μ and σ describe the population distribution.
They are **not** calculated from the sample unless we use estimates such as $\bar{x}$ and s.

The sample mean as a random variable

Let

$$X_1, \ldots, X_n$$

be the observations from a random sample.
Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Because the X_i depend on which units are selected, $\bar{X}$ also depends on the selected sample.
Therefore:

$\bar{X}$ is a random variable.

After the data are collected:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

is the observed value of that random variable.

A virtual sampling experiment

Let's imagine that the forest contains 100 000 trees.
We repeatedly select samples of 20 trees.
For each sample we calculate $\bar{x}$.
In R, conceptually:

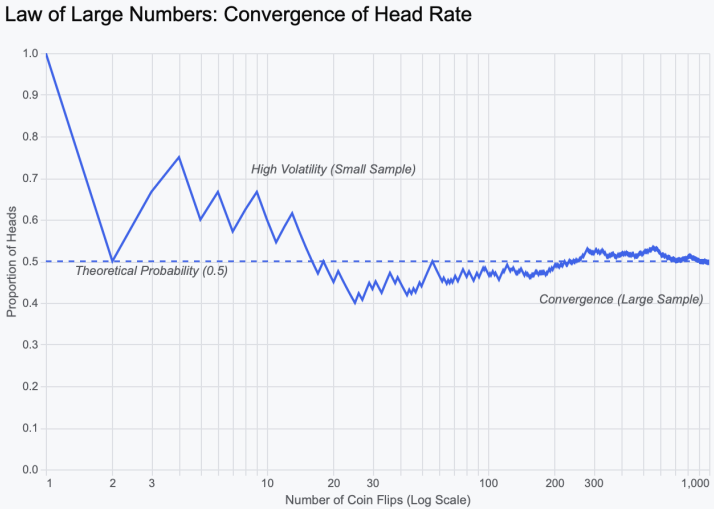
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repeat 1000 times:
    sample 20 trees
    calculate their mean diameter
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The 1000 means form an empirical approximation to the sampling distribution of $\bar{X}$.

This is a powerful idea

We can **see** the random variable $\bar{X}$ by repeating the sampling experiment.

The sampling distribution of $\bar{X}$



The mean of the sampling distribution

If the observations are independent and identically distributed with

$$E(X_i) = \mu,$$

then

$$E(\bar{X}) = \mu.$$

So, over repeated samples, the sample mean is centred on the population mean.

$$\boxed{\text{population mean} = \text{mean of the sampling distribution of } \bar{X}}$$

This is why $\bar{X}$ can be used to estimate μ.

Why does a larger sample help?

If

$$\text{Var}(X_i) = \sigma^2$$

and the observations are independent, then

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Therefore

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

$$n = 10 \quad \longrightarrow \quad n = 50 \quad \longrightarrow \quad n = 200$$

wide sampling distribution $\longrightarrow$ narrow sampling distribution

Interpretation

Larger samples make the sample mean more stable from one sample to another.

The Law of Large Numbers

As the sample size increases,

$$\bar{X}_n \longrightarrow \mu$$

under suitable conditions.
This is the Law of Large Numbers.

Intuition

With more and more observations, the average becomes increasingly representative of the population mean.

But there is an important distinction:

Law of Large Numbers: where does $\bar{X}$ go?

versus

Sampling distribution: how variable is $\bar{X}$?

The Central Limit Theorem

The Central Limit Theorem gives another crucial result.
Under suitable conditions, for sufficiently large n,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1).$$

This means that the sampling distribution of the mean is often approximately normal, even when the individual observations are not perfectly normal.

Do not confuse the two

The **individual observations** need not be normal for the **sample mean** to be approximately normal.

This is one reason the normal distribution is so important in statistical inference.
Two mathematical theorems connect probabilities with frequencies:

- The Law of Large Numbers says that, in the long run, the probability of an event equals the frequency of its occurrence;
- The Central Limit Theorem says that probabilities for the sum or average of a decent

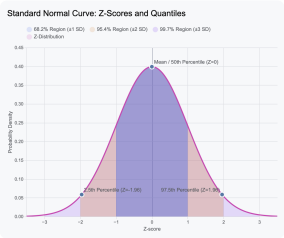
The Normal distribution

The standard normal distribution is

$$Z \sim N(0, 1)$$

with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$



For our purposes, the important properties are:

- centre $= 0$;
- standard deviation $= 1$;

Standardization

If

$$X \sim N(\mu, \sigma^2),$$

then

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

The same idea applies to the sample mean:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}.$$

This tells us how many standard errors $\bar{X}$ is from μ.

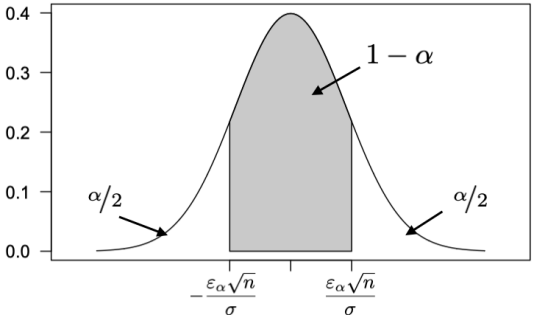
$$\boxed{\text{distance from } \mu \div \text{sampling variability} = \text{standardized distance}}$$

Quantiles and probabilities

Let $\Phi(z) = P(Z \leq z)$ be the cumulative distribution function of the standard normal. The p-quantile z_p satisfies

$$P(Z \leq z_p) = p.$$

For example: $z_{0.975} \approx 1.96$.
Therefore $P(-1.96 \leq Z \leq 1.96) \approx 0.95$.



From a probability statement to an interval

Suppose

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1).$$

Then

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95.$$

Now rearrange the inequality:

$$P\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95.$$

This is the key transition

We have transformed a probability statement about the random variable $\bar{X}$ into a procedure for constructing an interval for the fixed parameter μ.

The confidence interval

When σ is known:

$$\boxed{\bar{x} \pm z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}}$$

For 95% confidence:

$$\boxed{\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}}$$

Notice the notation:

$\bar{X}$ in the probability statement, but $\bar{x}$ in the final interval.

The interval is calculated from the **observed sample**, but its statistical justification comes from the **sampling distribution**.

Why the confidence interval has a width

The margin of error is

$$z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

It becomes larger when:

- the population is more variable;
- we demand a higher confidence level;

and smaller when:

- the sample size increases.

$$\boxed{\text{uncertainty} \sim \frac{\text{variability}}{\sqrt{\text{sample size}}}}$$

But usually σ is unknown

In real research, σ is usually unknown.
We estimate it with

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}.$$

For a normal population,

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}.$$

Therefore:

$$\boxed{\bar{x} \pm t_{n-1,1-\alpha/2}\frac{s}{\sqrt{n}}}$$

The t distribution accounts for the extra uncertainty from estimating σ.

A numerical example: tree diameter

Suppose we measure

$$n = 25$$

trees and obtain

$$\bar{x} = 31.4 \text{ cm}, \qquad s = 5.0 \text{ cm}.$$

A 95% confidence interval is

$$31.4 \pm t_{24,0.975} \frac{5.0}{\sqrt{25}}.$$

Since

$$t_{24,0.975} \approx 2.064,$$

the margin of error is about

$$2.06 \text{ cm}.$$

Thus:

$$\boxed{[29.3,\ 33.5] \text{ cm}}$$

Interpretation

What does “95% confidence” mean?

Imagine repeating the entire sampling procedure many times.
Each sample gives a different interval:

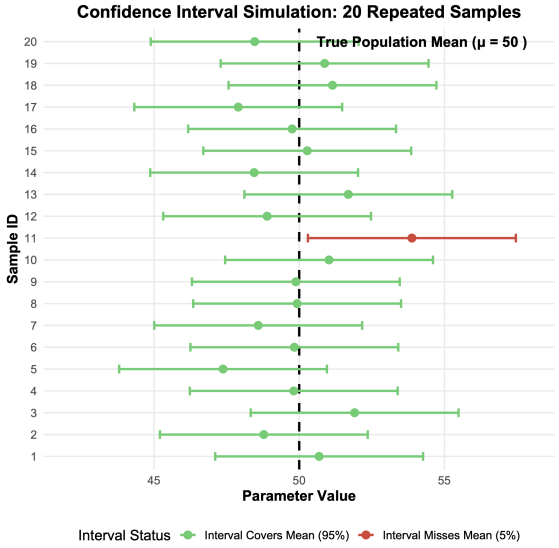
$$I_1, I_2, I_3, \ldots, I_{1000}.$$

The population mean μ stays fixed.
The intervals change because the samples change.
Approximately 95% of the intervals will contain μ if the procedure and assumptions are appropriate.

The correct mental picture

The parameter is fixed; the interval is random before the sample is observed.

What does “95% confidence” mean?



The three levels students must keep separate

Level	Example	Status
Population	μ	fixed, usually unknown
Sample	$x_1, \ldots, x_n$	observed data
Statistic	$\bar{x}$	calculated from the sample
Random statistic	$\bar{X}$	variable before sampling

The central distinction

$\bar{X}$ is random because the sample is random; $\bar{x}$ is the value obtained after sampling.

A useful mental experiment

Ask yourself:

"If I repeated the sampling procedure, would I obtain exactly the same $\bar{x}$?"

If the answer is **no**, then before sampling:

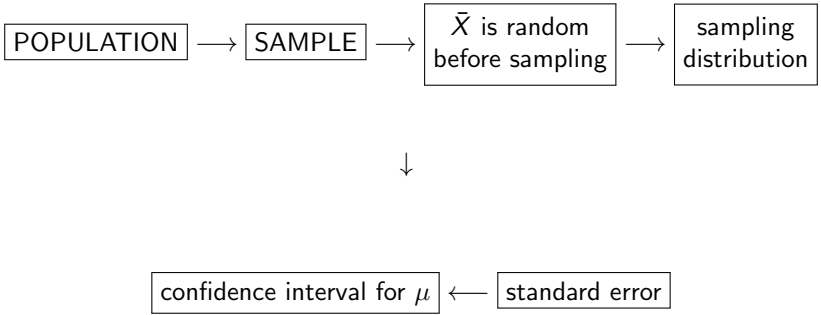
$$\bar{X} \text{ is random.}$$

After one particular sample:

$$\bar{X} = \bar{x}.$$

This simple thought experiment is often more useful than memorizing the definition of a random variable.

From population to confidence interval



What you should take away

1. A **population** is the complete set of units about which we want to make a statement.
2. A **sample** is the set of units actually observed.
3. A population parameter such as μ is fixed; the sample changes from one sampling operation to another.
4. A statistic calculated from a random sample is a **random variable before sampling**.
5. $\bar{X}$ is the random sample mean; $\bar{x}$ is its observed value.
6. The **sampling distribution** describes how a statistic varies across repeated samples.
7. The standard error quantifies the variability of a statistic.
8. Confidence intervals use the sampling distribution to quantify uncertainty about a population parameter.

Question for discussion

Two researchers study the same forest.
Both obtain

$$\bar{x} = 31.4 \text{ cm}.$$

Researcher A measured 25 trees.
Researcher B measured 400 trees.

Should they have the same confidence interval? Why?

Think first in terms of the **sampling distribution of** $\bar{X}$, not the formula.

What changes when the sample becomes larger?